essential parameter was disclosed and, thus, the skilled person was not able to make any measurement of the relevant essential parameter (decision cited in **T 206/08** (detergent compositions) but found inapplicable to the case in hand).

6.6.4 Forbidden area of the claims

As today there is a clearly predominant opinion among the boards that the definition of the "forbidden area" of a claim should not be considered as a matter related to Art. 83 and 100(b) EPC (**T 1811/13**, **T 647/15**, **T 646/13**).

In other words, according to established case law of the boards of appeal, the question of whether a skilled person is working or not within the claimed scope ("forbidden area" of a claim) is related to the definition of the scope of protection sought (Art. 84 EPC) and not to the sufficiency of disclosure of the invention (Art. 83 EPC) (as summed up in **T 1673/15**). See also Case Law of the Board of Appeal, 9th edition, 2019, for an overview of the earlier – or now at least minority – case law taking a different view.

See also chapter II.C.8.2. "Article 83 EPC and clarity of claims".

6.6.5 Non-disclosed steps

There is no requirement in the EPC that the claimed invention may be carried out with the aid of only a few additional non-disclosed steps. The only essential requirement is that each of those additional steps be so apparent to the skilled person that, in the light of his common general knowledge, a detailed description of them is superfluous (**T 721/89**).

6.6.6 Machine not available

In **T 1293/13** the claims limited the determination of air permeability of the garment to a particular method and to a particular machine ("Frazier 750"). However, the machine "Frazier 750" no longer existed, such that this machine could not be used for determination of the claimed values. The insertion of a feature defined as determinable by a specific machine which possibly was not – but certainly is no longer – publicly available, leads in this case to the invention not being disclosed in a manner sufficiently clear and complete for it to be carried out by a person skilled in the art.

In **T 1714/15** the board noted that the DROP-WEIGHT TESTER RTD-5000, as described in claim 1, was uncontestedly never available. Rather, the known apparatus was a DROP-WEIGHT TESTER RDT-5000 (emphasis by the board). The respondent (patent proprietor) argued that it had provided evidence that the apparatus with the correct designation was available at the priority date and at the date of filing of the patent in suit. Thus the invention as defined in claim 1 was enabled at that point in time, which the board accepted. However, the board stated that an invention had to be enabled throughout the **whole lifetime** of a patent (**T 1293/13**). This was not the case here. The respondent stated that there was no proof that the apparatus recited in claim 1, with the correct designation, did not exist somewhere in the world. However, the respondent had stated in its reply to the grounds of appeal that the contentious apparatus had become unavailable. In such a case, following the principle of "negativa non sunt probanda" (**T 2037/18**), the burden of proof that the apparatus had ceased to exist was not on the opponent. Rather, the proprietor had to prove that such apparatus was still available. In the absence of such proof the board concluded that the apparatus was no longer available, (requirements of Art. 83 EPC not met).

6.6.7 Experiments

The requirement of sufficient disclosure is intended to ensure that the skilled person can carry out the invention without having to do their own research or performing an unreasonable number of experiments. It is not met if they still cannot successfully carry it out after a few instructive attempts and, instead, must first employ their own inventiveness (**T 312/88**, point 3.3 of the Reasons; **T 516/99**, point 3 of the Reasons).

The legislative purpose of Art. 83 EPC is to ensure that the skilled person can reproduce the invention without his own research or undue experimentation. Experiments are an undue burden if their primary aim is to find the solution to the problem but not if they are carried out merely to determine the numerical limits of a functionally defined range (**T 312/88**). They should quickly give a reliable picture of how the products can be produced or manufactured (**T 475/88**). However, it is not necessary for the experimental data filed with the patent in suit to be an exact repetition of the worked examples of the patent, as long as the experimental work can be regarded as being within the scope of the invention (**T 674/96**).

For experiments to be considered reasonable, the application need not disclose the best and easiest method; a long and complicated route which is nevertheless clearly successful may be considered reasonable (**T 412/93**).

The board in **T 1133/08**, faced with a multitude of options for selecting suitable materials, dimensions and procedural parameters which were merely outlined in the part of description relating to embodiments, found that there was no specific information describing in detail at least one way of carrying out the invention claimed. Experiments were needed to solve the problem (i.e. identify parameters and conditions resulting in a sinusoidal profile), and, as established in **T 312/88** in conjunction with **T 68/85** and **T 18/89**, such experimentation had to be considered unduly burdensome. If an invention had several variants, it was very important first of all to describe as many as possible in detail, rather than merely outlining them, to show the skilled person that the invention could be carried out in practice across the entire breadth of the claims. Here, not one single way of carrying out the invention was apparent, nor had any subsequently been demonstrated, e.g. on the basis of experiments. The board also analysed **T 14/83**, contrasting it with **T 412/93** (genetic engineering).

In **T 345/09** (method for manufacturing parts with very high mechanical properties) the board found that the skilled person, faced with a lack of relevant examples relating to the invention's essential mechanical features, would have had to carry out a number of tests in order to arrive at it. Given the number of mechanical features (at least eight) and parameters (fourteen), combined with the ranges to be applied to these parameters and the repetitions of tests – needed for some mechanical features having to be guaranteed before and after quenching – all of which were essential to carry out the invention, the research programme facing the skilled person was, the board concluded, so extensive that it amounted to an undue burden.

6.6.8 Calibration and identifiable measurement method

In **T 641/07** (measurement method could be identified by reproducing an example), the main issues the board had to deal with were whether measurement methods could be identified, whether an incorrect reference to a measurement method could be corrected and whether details of how to perform the method of measuring one of the parameters could be ascertained. The board held, citing **T 485/00**, that when a skilled person was enabled to reproduce the invention, and it was sufficient for him to reproduce one of the examples in order to identify the method employed to measure the value of a parameter, there was no insufficiency in the description since the identification procedure in question could not be regarded as involving an undue burden.

The board in **T 1224/15** was, in particular, unconvinced by the argument that measuring the level of a polyamide's crystallinity was insufficiently described. The documents produced proved that this feature and the method of measuring it were among the skilled person's capabilities as at the invention's date of priority. The skilled person was therefore capable of supplementing the instructions given in the patent on how to measure the relevant parameters. The patent also contained a number of examples that were detailed enough to enable the skilled person to reproduce some of them and so identify or calibrate the measurement method used (see **T 641/07**). Lastly, the argument was not supported by any tangible evidence or (failed) attempt to reproduce the examples. The board was thus unconvinced that there was any lack of information in the patent that prevented the invention claimed from being carried out.

In **T 786/15** the board considered that, since it was possible for the skilled person to calibrate the method of measurement of the Tg by reverse engineering, the Tg parameter recited in claim 1 was not ambiguous, even at values close to the end ranges. Art. 83 EPC was fulfilled in this case.

For other decisions dealing with the calibration of a method in the context of Art. 83 EPC, see **T 1712/09** (no attempt at calibration made – burden of proof), **T 1062/98** and **T 485/00** (possibility of calibrating methods of determining the relevant parameters), and **T 45/09**.

6.6.9 Analytical measuring methods

Where it is obvious that a skilled person would select a particular analytical measuring method, (none being disclosed in the patent), balancing its simplicity and convenience against the required accuracy, the requirements of Art. 83 EPC are met (see e.g. **T 492/92**). This is the case even if the two different analytical methods proposed by the patentee give significantly different results with the same composition. It also suffices if the person skilled in the art would assume that it was most likely that a certain method was used and this assumption could be tested in the light of the information given in the examples of the patent in suit (**T 143/02**). However, where there are different measuring methods which do not always lead to the same result, this can amount to an undue burden, as in **T 225/93**. In **T 930/99**, the board considered **T 225/93** inapplicable, as there was only one measurement method before them. The respondent's argument that there would be legal uncertainty, since third parties would not know whether they were working within or outside the range specified, was clearly an argument based on lack of clarity, which was not a ground of opposition and so could not be considered (see also in this chapter II.C.8.2.).

See also chapter II.C.5.5.1 "Ambiguous parameters", where the issue of methods of determining parameters is likewise addressed.

6.6.10 Chemical compounds

According to **T 954/05**, the structural definition of a chemical compound may not be replaced in a claim by the mere juxtaposition of a feature purportedly representing a complete chemical structure and of a functional feature if on the one hand the first feature comprises an indefinite number of compounds and there is no systematic selection rule based on the feature in question enabling the skilled person to identify the claimed compounds, and on the other hand the second, functional feature is not identifiable in the indefinite list of compounds potentially suitable for such a function because there is no indication of a typical standardised test for determining its presence or absence.

In **T 544/12** the board confirmed that a definition of a group of compounds in a claim by both structural and functional features is generally acceptable under Art. 83 EPC as long as the skilled person is able to identify, without undue burden, those compounds out of the host of compounds defined by the structural feature(s) in the claim which also fulfil the claimed functional requirements (following **T 435/91**, OJ 1995, 188, and **T 1063/06**). In **T 544/12** it was up to the skilled person to identify within the almost infinite host of alternatives covered by the structural definition of claim 1 those compounds that were phosphorescent. Claim 1 extended to classes (of iridium complexes) that were entirely different from the concept as argued by the proprietor (non-compliance with Art. 83 EPC). The board did not share the view taken by the German Federal Court of Justice (Bundesgerichtshof) in its decision of 11 September 2013 (X ZB 8/12).

See also **T 959/08** (requirement that a functional feature be implemented also highly relevant for Art. 83 EPC – no concept fit for generalisation – undue burden) and chapter II.A.3.4 "Functional features".

The very detailed decision **T 842/14** concerned a chemical composition of a product designated by a **trademark** (see also **T 270/11** and **T 623/91**). According to **T 667/94**, **T 325/13**, and **T 1383/10**, when the products designated by trademarks are essential for carrying out the invention, the requirements of Art. 83 EPC are fulfilled if these products are available to the skilled person not only at the priority and filing dates of the patent but also during its **whole lifetime** (in **T 842/14** no certainty that the composition would remain unchanged). In this respect **T 842/14** contains extensive reasoning on the distinction between the requirements of Art. 83 EPC and those of Art. 54 EPC (especially in view of **G 1/92**, OJ 1993, 277).

6.7. Trial and error

Even though a reasonable amount of trial and error is permissible when it comes to sufficiency of disclosure, e.g. in an unexplored field or where there are many technical difficulties, the skilled person has to have at his disposal, either in the specification or on the basis of common general knowledge, adequate information leading necessarily and directly towards success through the evaluation of initial failures (**T 226/85**, OJ 1988, 336; following **T 14/83**, OJ 1984, 105; **T 48/85**, **T 307/86** and **T 326/04**; see also **T 2220/14**, highly complex technical field). Where the skilled person can only establish by trial and error whether or not his particular choice of numerous parameters will provide a satisfactory result, this amounts to an undue burden (**T 32/85**; see also **T 2237/09**, concerning a combination of parameters). Nor can sufficiency of disclosure be acknowledged, if, for an invention which goes against prevailing technical opinion, the patentee fails to give even a single reproducible example (**T 792/00**. See also **T 397/02**, **T 1440/07** and **T 623/08**).

In case **T 2220/14**, in the board's view, since the technical field to which the invention related was highly complex (methods of modifying eukaryotic cells), the average amount of effort necessary to put a written disclosure into practice in this field would be rather high and involve a considerable amount of trial and error. The board added that there is no requirement in the EPC, either at the priority or filing date, that the applicant must have carried out the claimed invention. The board concluded that it had no reason to doubt that the invention as claimed in claims 1, 5 and 6 was sufficiently disclosed as required by Art. 83 EPC.

Where the person skilled in the art has to find out by trial and error which, if any, compound meets the parameter set out in the claim, this constitutes an undue burden. The fact that this could be done by routine experimentation was not sufficient for the subject-matter claimed to meet the requirements of Art. 83 EPC. Nor did the question whether or not the parameter could be reliably determined play a role (**T 339/05**). In **T 123/06** the board found that the **functional definition** of the device was no more than an invitation to perform a research programme, the skilled person only being able to establish through trial and error whether the claimed device was achieved. This amounted to an undue burden.

According to **T 1063/06** (OJ 2009, 516), a functional definition of a chemical compound (in this case in a **reach-through claim**) covered all compounds possessing the capability according to the claim. In the absence of any selection rule in the application in suit, the skilled person, without the possibility of having recourse to his common general knowledge, had to resort to trial-and-error experimentation on arbitrarily selected chemical compounds to establish whether they possessed the capability according to the claim; this represented for the skilled person an invitation to perform a **research programme** and thus an undue effort (following **T 435/91**). See also **T 1140/06**.